

University of Verona

DEPARTMENT OF COMPUTER SCIENCE
Master's degree in Data Science

Dynamic User Profile Injection: A Closed-Loop RAG Architecture for Context-Aware LLMs

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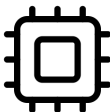
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Context & The Problem



The Context: European DataGEMS project & MathE EdTech platform.



The Technical Bottleneck: Foundational LLMs are inherently *stateless* (no episodic memory).



The Pedagogical Flaw: Standard models cannot track a user's historical misconceptions or temporal memory degradation.

Data Exploration & Sparsity



Source: Assessment_Information table (20,241 interaction logs) from MathE platform.



Feature Selection:

- Filtered meta-data (Typology, question_id) to focus on cognitive signals.
- Behavioral metrics (duration, option_selected) discarded due to extreme sparsity (only 30 non-null values).



Hierarchical Imputation: Resolved ~31% null rate in Subtopic using Topic labels

Column	Non-Null Count	Dtype
student_id	20,241 non-null	int64
question_id	20,241 non-null	int64
answer	20,241 non-null	int64
date	20,241 non-null	object
duration	30 non-null	float64
option_selected	30 non-null	float64
Topic	20,241 non-null	object
Subtopic	13,803 non-null	object
Lecturer_level	20,241 non-null	int64
Algorithm_level	20,241 non-null	int64
Typology	20,241 non-null	object

Dataset Construction - I



Data Aggregation:

- User Selection: Isolated the most active/temporally consistent student profile.
- Event-Level Metrics:
 - Knowledge (K): $K_{event} = (m \cdot d) \cdot W_k$
 - Lapse (L): $L(\Delta t) = 100 \cdot e^{-\frac{\max(\Delta t, 0)}{S}}$
 - Interest (I): Dynamically computed by the system's Evaluator based on conversational engagement.
- Macro-Aggregation: Grouped by Topic & Subtopic to form the State-Tracking Vector.

Dataset Construction - II



Synthetic Expansion: Synthetic records generated to stress-test the architecture



Final Dataset: >1,000 unique longitudinal records for high-fidelity state tracking.



Semantic Discretization

- Strategy: Mapping continuous scores (0-100) into 5-tier pedagogical labels.
- Labels: Extremely Low to Extremely High (for K , L , and I).
- Result: A distilled State-Tracking Vector injected into the system prompt.

Dataset Construction - III

Topic	Subtopic	L_sc	K_sc	I_sc	L_cat	K_cat	I_cat	last_int_date
Analytic Geometry	Analytic Geometry	32.0	17.0	18.0	Highly lapsed	Ex. low knowledge	Ex. low interest	2026-05-30
Complex Numbers	Complex Numbers	2.0	40.0	18.0	Ex. lapsed	Low knowledge	Ex. low interest	2026-05-29
Differential Equations	Differential Equations	91.0	22.0	50.0	Not lapsed	Low knowledge	Moderate interest	2026-05-31
Differentiation	Derivatives	87.0	65.0	92.0	Not lapsed	High knowledge	Ex. high interest	2026-05-31
Differentiation	Differentiation	27.0	39.0	18.0	Highly lapsed	Low knowledge	Ex. low interest	2026-05-30

The Architectural Triad: Shared Functional Pipeline



The Core Pipeline: Every iteration relies on the same tri-layers functional loop:

01. *Router:* Context triage and semantic extraction.
02. *Generator LLM:* Proactive, context-grounded pedagogical tutoring.
03. *Asynchronous Evaluator:* User profile update layer via LLM-as-a-Judge.

The Architectural Evolution: From Naive to Routed

Naive RAG (The Baseline)



Logic: Brute-force injection of the entire database.

Constraint: No filtering; subject to "Lost in the Middle" syndrome.

Flat Router (FRB)



Logic: Probabilistic matching against a linear, flat topic list.

Constraint: High risk of "Keyword Traps" and semantic over-extraction.

Hierarchical Router (HRB)



Logic: Deterministic traversal of a parent-child topological map (Topic + Subtopic).

Constraint: Surgical retrieval, higher routing-layer token overhead.

The Engineering Trade-off: FRB vs HRB

The FRB Trade-off (Cost-Efficiency)



Pros: Minimal routing overhead (748 tokens). Best for simple, explicit queries.

Cons: Prone to "Shotgun Extraction"—pulling irrelevant nodes to compensate for ambiguity.

The HRB Trade-off (Structural Precision)



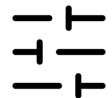
Pros: Deterministic traversal ensures surgical, noise-free context.

Cons: Higher initial routing cost (9.4k tokens). A deliberate investment for semantic integrity.

The Generator (Proactive Pedagogical Tutor)



Role: Generative layer powered by gemini-2.5-flash



Pedagogical Modulation:

- **Cognitive Alignment:** Dynamically adjusts learning paths and interventions by mapping Knowledge and Lapse tiers to targeted difficulty levels, ranging from refresher training to advanced practice.
- **Affective Alignment:** Modulates tone and engagement strategies based on Interest scores (e.g., Real-world applications vs. Sparking curiosity).

The Asynchronous Evaluator



Role: Background user-profile updater layer powered by gemini-2.5-flash-lite.

Core Functions:

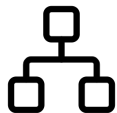


- **State Extraction and Update Context:** Maps raw dialogue into quantified K and I metrics and apply EMA to prevent profile volatility from single anomalous events.
- **Semantic Bypass ("Kill Switch"):** Filters non-pedagogical meta-intents (e.g., travel plans) to protect database integrity.
- **Open Ontology Strategy:** Employs Greedy Matching with dynamic expansion for Out-of-Domain (OOD) topics.

The Closed-Loop HRB Architecture



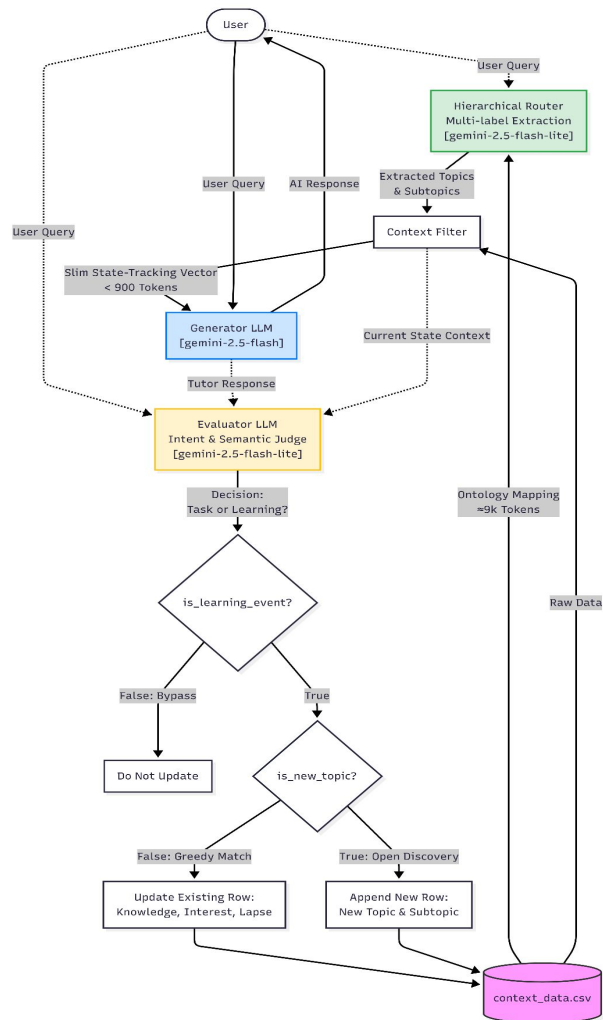
Hierarchical Router: Deterministic multi-label extraction (prevents over-extraction).



Generator LLM: Proactive context-aware pedagogical tutoring.



Asynchronous Evaluator: Background user-profile updater layer.



Interaction Case Study: Closed-Loop in Action

User Query

User: "I keep failing the mock tests on organic chemistry nomenclature. I just don't get the IUPAC rules and I'm getting really tired of it."



Router Extraction

```
{  
  "topic": "Chemistry and Biochemistry",  
  "subtopic": "Organic Chemistry and Drug Synthesis",  
  "lapse_cat": "Moderately lapsed",  
  "knowledge_cat": "Extremely low knowledge",  
  "interest_cat": "Extremely low interest"  
}
```



Context_Data



Context update

```
{  
  "is_valid_tracking_event": true,  
  "topic": "Chemistry and Biochemistry",  
  "subtopic": "Organic Chemistry and Drug Synthesis",  
  "interaction_knowledge": 12.0,  
  "interaction_interest": 8.0  
}
```

System Action: Update existing row using EMA

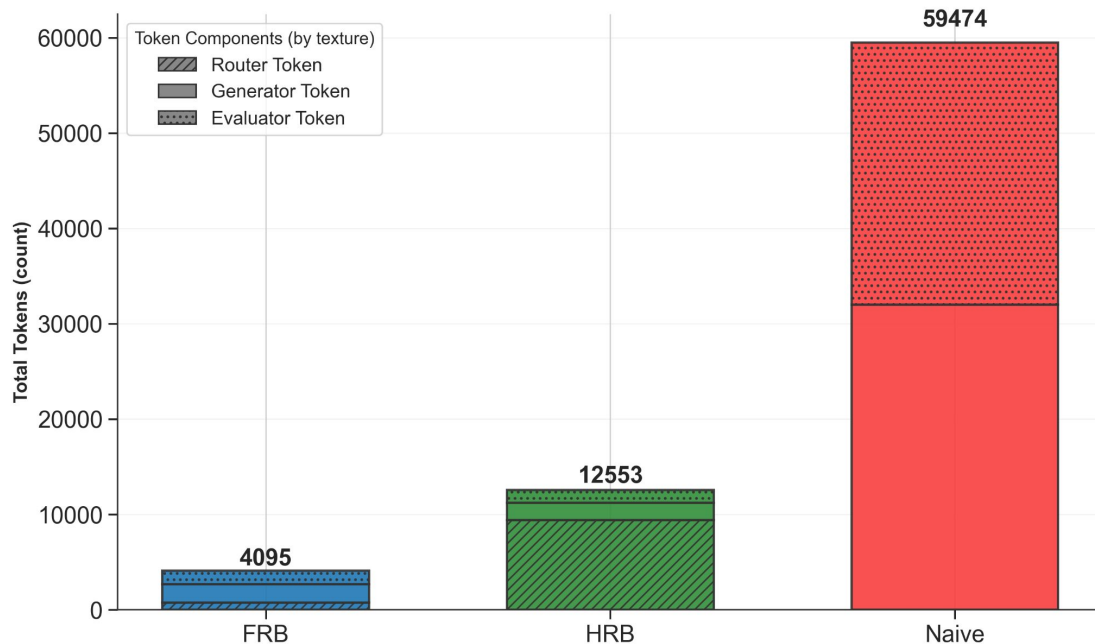


Generator Response

Generator: [Offers empathetic support + foundational review of IUPAC rules].

Empirical Results & Conclusions I

↙
↗ **Context Compression:** ~79.2% reduction in e2e token usage (Naive vs HRB).



Empirical Results & Conclusions II



Latency Optimization: Generation time stabilized at ~6 seconds, with background evaluation executing in ~1 second.



The "Closed-Loop" Proof of Concept: Successfully tracked temporal memory decay without explicit questionnaires. Maintained structural database integrity via active Semantic Bypass.

Limitations & Future Directions

Current Limitations



- The Global Profile Query Dilemma: Local routers struggle with holistic meta-queries.
- API Dependency & Cost: High routing overhead (9.4k tokens) per turn poses scalability challenges.

Future Directions



- Hybrid Intent Classification: Implementing upstream semantic switches for meta-queries.
- Dense Vector Retrieval: Transitioning from static hierarchical maps to Vector Databases (e.g., Pinecone) for multi-hop semantic search.

THANKS FOR YOUR ATTENTION

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